

[Request] What uses more water and energy, a human sitting at their computer and crafting a CGI image over the course of 8 hours, or an AI model that generates a similar image in 5 minutes?

Posted by u/Cixin97 • • 0 points • 31 comments

I'm posting this in lieu of the constant hatred towards AI usage on Reddit. Last night someone posted a hypothetical situation for a movie franchise I like and included an AI generated image to go with his description. This set of what can essentially be described as a brigade. All of the comments were about it being AI slop (frankly it looked cool though), and things like "how much water did the server guzzle to create this image?". Sentiments like that are becoming more and more common on reddit and even though this post might be better off without me giving my perspective, I will anyone. I think comments like this are bs, anti advancement, Luddite, etc. More importantly they're not acknowledging Jevon's paradox (Jevons paradox, also known as the Jevons effect, is an economic concept which occurs when technological improvements increase the efficiency of using a resource, thereby reducing the amount needed per application. However, as the cost of using the resource drops, if demand is price elastic, the overall demand for the resource actually increases, causing total consumption to rise rather than fall) meaning many of these people are upset that OP used AI rather than paying an artist to make a similar image. The reality is that the vast vast majority of people posting AI generating images would not have ever considered using a human artist to make the same image. It would have cost at least 10x more (likely more like 100x more to get a similar quality image), would have been logistically far more complicated (at best finding an artist on Fiverr or similar, paying them, setting timelines, waiting, wondering if they're actually going to deliver, etc), and the feedback loop for adjusting image would be far slower. I can almost guarantee that in the example above OP simply would've posted the description without an image if AI generated images weren't an option. So he didn't take any work from a real artist.

But to the main part of the post, can anyone do some napkin math on how much energy and water one of these models likely burns to create an image and compare that to a human?

If you'd like, feel free to be extremely conservative towards either side of the equation. For example if you want to favour humans, don't account for the fact that the human themselves requires food/water/energy to survive, don't account for the fact that they need a monitor to view their work (which an AI model does not need, etc). On the flip side if you want to favour

AI, consider the fact that they're using servers in places highly optimized for electricity usage, water throughout, etc. It's also worth noting that most models that are at scale are now using very specialized hardware that is far more efficient than an off the shelf CPU and GPU for doing said work. You could account for that.

I was also fair in the title about timespans imo, but you can adjust that if you think it's unfair. I do modelling as a hobby and also play around with these models and I believe 8 hour human work to 5 minute AI model work is beyond fair. The reality is it would likely be more like 1 minute worth of a model running to get a similar quality image (to the layperson) for 8 hours of human work.

But in general I'd like a relatively balanced approach. My intuition is that in reality a human on their computer probably burns 10x more energy and water than an AI at minimum, and it wouldn't surprise me if it's more like 100x.

Comments

[u/\[deleted\]](#) • 1 points • 2025-12-02 06:59 UTC

[removed]

[u/DJSauvage](#) • 2 points • 2025-12-02 03:09 UTC

The ease of creating AI slop means that for every image used and seen there's many that were created and discarded. Also, there is still a person prompting the AI, also using energy and water, and the cost of training the AI, and running it computationally.

[u/muhbalwzishawt](#) • 3 points • 2025-12-02 00:15 UTC

How do you account for the fact that AI allows millions more people choose to "generate CGI" who would otherwise choose not to?

This argument folds to perspective of scale.

[u/Cixin97](#) • 0 points • 2025-12-02 07:32 UTC

That's not a bad thing. That's like saying "cars are great because they allow people to go far further for different opportunities or to visit family more often" and someone says "well actually that's bad because more people are driving than if cars didn't exist".

[u/muhbalwzishawt](#) • 1 points • 2025-12-02 13:03 UTC

More AI use is objectively bad.

[u/Cixin97](#) • 0 points • 2025-12-02 19:24 UTC

No it's not, and I don't think you know what objectively means.

[u/III_Barber8709](#) • 2 points • 2025-12-01 22:43 UTC

I'm a die-hard ecologist and I too think this argument is dumb as fuck.

I'm also a daily user of local LLMs, a developer and a gamer.

Generating tokens (the thing that generative AI produces, being text, sound, images or whatever) for the time you get your answer won't be more energy-consuming than said time playing a video game. That's it.

The problem behind AI servers is not in **AI** but in **servers**. It's the result of centralised computing, which locally consumes a lot of energy and produces a lot of heat that is lost in the air in most cases.

So sure, AI servers are generating pollution when they're using CO2-emitting energy. And they add up to the existing demand. But pointing at AI as the mother of all evil doesn't help solving the true issue, which is our increasing dependence on centralised compute.

I do think AI has a destructive potential for society as we know it though, and we must be extra careful about it. Think of nuclear energy kind of threat. It could bring incredible progress for all, as well as put us all out of work.

TL;DR: GeForce Now users suck too. Buy local.

u/Cixin97 • 1 points • 2025-12-01 22:51 UTC

I agree there are massive risks (potentially even greater than nuclear) but isn't even your outlook on things downplaying the economies of scale involved with having computer done all in one location vs locally? These data centres are highly optimized in terms of electrical usage, heat waste, water usage, etc. They're also specifically placed in areas that have high electricity availability and many of them are placed in places where the tech companies are hoping to expand clean energy (see Microsoft nuclear deals) to run the data centre's. Compare that to my computer. Minimal thought into heat waste and energy usage beyond how what FPS it allows my game to hit by keeping computer semi cool. I don't really care how much electricity it uses or where it comes from because even if it's 2x what it would be by being optimized, that 2x increase/decrease is still a drop in the bucket compared to my overall energy bill, and that % of my energy bill is not even remotely coming close to dictating where I'd want my house to be in order to get cheaper electricity. For data centres those are actual considerations. For me, even if I do 8 hours of gaming and 3D modelling per day it's the difference between my bill being \$20 per month vs \$10 per month, and that's assuming very heavy loaded for the entire duration of 8 hours. More accurately it would be the difference between \$10 per month vs \$5 per month. Not exactly the most pressing issue.

u/III_Barber8709 • 1 points • 2025-12-01 23:23 UTC

Computer chips (CPU, GPU, RAM, Flash, whatever) basically are heat sinks. The amount of energy necessary to produce the information^[^1] they produce (the bits) is totally negligible compared to the amount of energy lost by electric resistance.

The best use we could make of the energy we put into our compute needs is heating something that needs to be heated near you. For example, I take a shower every day.

It's true that everything is ultra-optimised in servers (well, at least some of them), but in the end, the heat produced is wasted, no matter how well the server is built.

[^1]: https://en.wikipedia.org/wiki/Information_theory

u/Cixin97 • 0 points • 2025-12-01 23:27 UTC

10-30% of that heat is reused at a data centre though. A negligible amount is reused at a home computer *unless* you're in a cold climate and it saves you from having to heat your space as much.

u/III_Barber8709 • 1 points • 2025-12-01 23:37 UTC

Even in hot climate you need to take hot showers. 10-30% is far from, and we have the possibility to reuse all of that heat. Building servers in space is the stupidest thing I've ever heard of. Build them in big cities, and sell hot showers.

u/Cixin97 • 1 points • 2025-12-01 23:43 UTC

Sure but again that's a point for centralized computing rather than decentralized. No one is ever going to attach a PC to their water tank. On the other hand doing that with a data center and having dedicated city hot water tanks is feasible even if still highly complex.

u/III_Barber8709
• 1 points
• 2025-12-01 23:57 UTC

> Sure but again that's a point for centralized computing rather than decentralized

I live in a cold climate, so my computer helps heating my house.

> No one is

ever
going to
attach a
PC to
their
water
tank.

You
mean a
*water
cooling*
tank? I
have a
feeling
we're
kinda
almost
there
already.

And I
would
definitely
buy an
electric
water
heater
PC if I
could.
That
would
be a
great
use of
that old
Radeon
currently
collecting
dust in
the attic,
don't
you
think?

> On
the
other
hand
doing
that with

a data center and having dedicated city hot water tanks is feasible even if still highly complex.

It is, but it's not currently how servers are built, while I can already heat my house generating tokens and playing games - instead of heating whatever server from the other side of the planet.

u/cha0sb1ade • 1 points • 2025-12-01 22:20 UTC

AI can generate the first garbage attempt in 5 minutes. But it's going to take lots of iterations to get to something usable. And still, it's probably going to have senseless artifacts that hint that it's just a cobbled together mess. This is very much an apple to oranges comparison you've set up. 5 minutes of AI does not produce output equivalent to 8 hours of skilled human labor. So while the math question could be answered, the premise is intellectually dishonest, and the answer is essentially meaningless

u/Cixin97 • -4 points • 2025-12-01 22:32 UTC

When was the last time you played around with AI image or video generators? I was being extremely favourable towards the human in my title. The reality is within the first try of an AI in 2025 you're more than likely going to get a result that is far better than the human would've generated in 8 hours, and the human who is capable of that level of work is actually going to be charging *far* more than 10x the price of the AI. A more accurate ballpark estimation would be that 3-5 attempts and refinements with AI would cost you <\$5, and the human achieving a similar result would cost \$100+ even if you use the cheapest artist possible from a third world country. I'm willing to bet you're in denial and haven't actually seen how good the models have become in 2025, and you're basing your opinion on 2023 models.

u/cha0sb1ade • 1 points • 2025-12-02 12:03 UTC

Hate to double reply, but something else that needs readdressed here. The energy cost of AI art isn't bests considered on a piece by piece basis anyway. Artists are not using the amount of electricity that AI art uses, overall, even though artists are where all the decent art is coming from. Mostly, AI is chugging away on requests that will never see the light of day. There are server farms full of virtual servers chugging away on requests like "4k anime style video of \[some famous celebrity\] but with \[some other color hair\] and 3 tits shaped like \[a famous porn star\]'s having sex with the hoodie wearing neckbeard from this photo I'm sending from my web cam. "

u/iaintevenueadcatch22 • 0 points • 2025-12-02 18:14 UTC

Also, costwise they're being subsidized by VC the same way amazon was subsidized for at least a decade to kill off the brick and mortar store

u/Cixin97 • 0 points • 2025-12-02 19:23 UTC

No, they're subsidized for future growth, not to kill of brick and mortar.

u/cha0sb1ade • 1 points • 2025-12-02 11:53 UTC

I work for a public institution with limited funds. We tried to hand our logos off to AI. I'm not even an artist. Just someone who took 8 hours of photoshop years ago. Still beat the AIs. You know why most people commission or do art? Because they want something very specific, not some approximation that describes what to do with 8 million pixels based on a paragraph. AI still delivers nonsensical geometry and shadows. Throws in weird artifacts that don't match the prompt because the source art it was trained on had something in that spot. It's pathetic.

Even if the quality problems were completely sorted out, the lack of specificity will still be an issue. Sometimes an approximation of a request might be okay, but most of the time on serious, business projects, you want pixel by pixel level control, reusing the exact colors shapes, items, characters clothes from different angles in different lighting, etc. If you're going to some one-off background image that doesn't need any consistency and won't get much scrutiny, AI can do that. Give it anything more complex, and you're in for frustration, getting yourself set up to look unserious. Besides all that, it's poor ethics.

TLDR: Comparing time spent on AI versus an artist to get 'similar results' is a nonsense question, because the artist is doing actual art, and the AI is doing something that equates to amalgamating existing works with very little precise control.